

Part 2 ACH Returns & ACH Events Grid (PAY-2342)

 [\[PAY-1803\] Automatic Retry of ACH Payments for Soft Return Codes \(R01, R09\)](#) Created: 14/May/25 Updated: 19/Jun/26

Resolved: 31/Oct/25

Status:	Closed
Project:	Payments
Components:	None
Affects versions:	None
Fix versions:	R8.27
Parent:	Part 2 ACH Returns & ACH Events Grid

Type:	Story	Priority:	3: Standard
Reporter:	Michelle Brahmabhatt	Assignee:	Brett Roy
Resolution:	Done		
Labels:	None		
Σ Remaining Estimate:	1 day, 6 hours, 15 minutes	Remaining Estimate:	1 day
Σ Time Spent:	4 weeks, 4 days, 1 hour	Time Spent:	0 minutes
Σ Original Estimate:	1 week, 3 days	Original estimate:	Not Specified

Attachments:	 20251016-qa.7z  21D044CF-6177-42DE-9D85-A20C82CF1202.jpeg  4338E475-45F6-4049-BA5B-F72285C2C279.jpeg  3E5FD0F0-E4CA-48DA-BDEA-6D178CF85C85.jpeg  E6CC45BE-57FF-429B-ACFA-43E7329003D4.jpeg  29bd3c9b-7149-4eee-9a97-be8fbfd1eec.jpg  09DC2E92-AF81-498A-BD2F-A2D48F925EFC.jpeg
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Issue links:	Relates relates to PLAT-3990 WFC - eChecks flag and retries upon f... Closed
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Sub-tasks:	Key	Summary	Type	Status	Assignee
	PAY-2313	QA: Create Test Cases	Sub-task	Done	Ana Oliva (contractor)
	PAY-2314	QA: Test in QA	Sub-task	Done	Ana Oliva (contractor)
	PAY-2316	Dev - Automatic Retry of ACH Payments	Sub-task	Done	Brett Roy
	PAY-2438	Release Config	Sub-task	Done	Brett Roy
	PAY-2441	Add publishEventGrid to AchReturns an...	Sub-task	Done	Brett Roy
	PAY-2494	Create CI/CD Pipeline	Sub-task	Done	Jason Thomas
	PAY-2555	Fix Tests in Treasury - Broken by ACH...	Sub-task	Done	Brett Roy
	PAY-2562	Deploy to QA	Sub-task	Done	Jason Thomas
	PAY-2710	QA: Help Ana	Sub-task	Done	Radhika Kandoori
	PAY-2730	Dev: Re-work	Sub-task	Done	Brett Roy
	PAY-2732	QA: Test Re-Work in QA	Sub-task	Done	Ana Oliva (contractor)
	PAY-2737	QA: Test in Stage	Sub-task	Done	Ana Oliva (contractor)
	PAY-2740	Dev: Setup Config / Re-Work	Sub-task	Done	Jason Thomas
	PAY-2767	QA: Re-test in Stage	Sub-task	Done	Ana Oliva (contractor)

Sprint:	PayR8.27 - S1, PayR8.28 - S1, PayR8.25 - S1, PayR8.26 - S2, PayR8.25 - S2, PayR8.26 - S1
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Acceptance Criteria:	AC1 – Trigger Retry for Eligible Return Codes If the merchant is configured for automatic ACH retries and a return is received with return code R01 (Insufficient Funds) or R09 (Uncollected Funds) , the system must automatically initiate a retry attempt. AC2 – Retry Attempt Limit The system must enforce a maximum of two (2) retry attempts per original ACH transaction (Within a month(30 -31 day window)). Any further retries beyond this limit must be prevented.
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AC3 - Re-deposit Retry via Worldpay: Each retry must use **Worldpay's redeposit endpoint**, which **does create a new ACH transaction**, but keeps it linked to the **original transaction**.

AC4- Maintain Traceability: The retry attempt must **reference the original transaction ID** for **reporting, audit, and reconciliation** purposes.

AC5 - Exclude Hard Return Codes*: Retry logic must **only apply to soft return codes** (R01, R09). Return codes like **R03, R10, etc.** must be excluded from retry logic.

Test Cases :

#	Test Case Description	Input	Expected Outcome	Validation Point
1	Retry triggered for eligible soft return code R01	Merchant is configured for ACH retry; return received with code R01	System initiates Retry 1 via Worldpay redeposit endpoint	Retry attempt logged; Event Grid updated; retry tied to original transaction
2	Retry triggered for eligible soft return code R09	Merchant is configured for ACH retry; return received with code R09	System initiates Retry 1 via redeposit	Same validations as Test Case #1
3	Retry not triggered for ineligible hard return code	Return received with code R03; merchant is configured	No retry is initiated	Event Grid shows only original return; system logs indicate retry not eligible
4	Retry not triggered when merchant not configured	Return code is R01; merchant retry flag is disabled	Retry logic is bypassed	System logs "Retry disabled for merchant"; no retry transaction
5	Enforce max retry attempts (2 retries max)	Original return + Retry 1 + Retry 2 completed; return received again	Retry 3 is blocked	System blocks retry ; Event Grid shows "Retry Limit reached"
6	Retry attempt is recorded under original transaction	Retry 1 initiated for R01 or R09	Retry is submitted via redeposit and is linked to original transaction ID	Retry record includes <code>original_transaction_id</code> in DB or API
7	Retry 1 is successful; no further retries	Retry 1 result = Success	Retry 2 is not triggered	System logs retry success; no new retry attempts queued
8	Retry 2 is triggered after Retry 1 fails	Retry 1 fails ; return code still eligible	Retry 2 is initiated via redeposit	Event Grid logs Retry 2; retry count incremented; linked to original transaction
9	Retry 3 is suppressed after Retry 2 fails	Retry 2 already executed ; return received again	Retry 3 is blocked	System logs indicate retry suppressed due to retry limit
10	Retry tracking for audit and reporting	Retry 1 or Retry 2 initiated	System records retry attempt with linkage to original transaction and outcome	Audit log entry created; DB/API contains reference fields for tracing

QA Required: Yes

Grooming Status: 7-Groomed

Pod: PAY

Description

As Wellfit,

I want the system to automatically retry failed ACH transactions that were returned due to soft return codes such as R01 (Insufficient Funds) or R09 (Uncollected Funds),

So that I can improve recovery rates, minimize financial risk, and ensure compliance with NACHA guidelines for re-presentment..

Business Context:

ACH payments occasionally fail due to soft return reasons like insufficient or uncollected funds. NACHA permits limited re-presentment of such transactions under specific conditions. Automating this retry process helps reduce manual intervention, recover legitimate funds more efficiently, and ensures consistent adherence to industry standards.

This capability supports key business goals:

- **Minimize Delays:** Respond to soft return codes in a timely and automated fashion.

- **Reduce Financial Risk:** Prevent payment losses in recurring or high-volume billing scenarios.
- **Maintain Compliance:** Stay aligned with NACHA rules for retry logic and consumer protection.
- **Improve Customer Experience:** Avoid over-crediting and proactively manage return fallout.

NOTE :

Retry logic should not apply for hard return codes (e.g., R03, R10).

Attaching the Retrun codes for reference :

http://support.worldpay.com/support/iq-reporting-and-analytics-user-guide/content/ach_return_reason_codes.htm#echeck_reason_codes_3170549809_723357

Retry eligibility is governed by configuration at the **merchant/sub-merchant level** within the **onboarding portal**. This includes:

- A flag to enable/disable automatic ACH retries
- A setting to define the **maximum retry attempts** per transaction (typically 1 or 2)

If a merchant has not enabled retry, or if the attempt limit is reached, the system must skip retry initiation regardless of the return code.

Comments

Comment by [Brett Roy](#) [11/Aug/25]

 [PAY-1598 - ACH Returns Management - Tech Design - Payments Space - Wellfit Wiki](#)

Comment by [Brett Roy](#) [12/Aug/25]

<http://support.worldpay.com/support/CNP-API/content/echecksale.htm>

In Chrome

Comment by [Brett Roy](#) [27/Aug/25]

On hold for a bit while ironing out some implementation details

Comment by [Brett Roy](#) [08/Sep/25]

[Ana Oliva \(contractor\)](#)

PRs have been submitted and waiting on Approval and deployment
I'll provide you testing files that I used once we get everything approved

Comment by [Jason Thomas](#) [11/Sep/25]

Quick notes: Happy path testing will require mocked data for QA to proceed.

Comment by [Jason Thomas](#) [14/Sep/25]

Opsie on the automation relating to drafting [Pull Request 14935](#): WF Payments – ACH Returns API

Comment by [Brett Roy](#) [18/Sep/25]

[Swetha Bayya](#)

AC2 – Retry Attempt Limit

The system must enforce a maximum of **two (2) retry attempts** per original ACH transaction (**Within a month(30 -31 day window)**). Any further retries beyond this limit must be prevented.

Is the 30-31 day window thing new? I do not recall seeing that prior, but I may have just missed it! 😊

Comment by [Swetha Bayya](#) [18/Sep/25]

[Brett Roy](#) This isn't new—I had included it when we initially groomed the story.”

Comment by [Jason Thomas](#) [24/Sep/25]

Awaiting deployment for testing

Comment by [Jason Thomas](#) [26/Sep/25]

Awaiting completion on [New CI/CD and PR Pipelines: ACH Returns API - DevOps and Infrastructure Support - Jira Service Management](#)

Comment by [Brett Roy](#) [06/Oct/25]

[Jason Thomas](#) Just confirming - we are still waiting, correct?

Comment by [Jason Thomas](#) [07/Oct/25]

[Brett Roy](#) This is correct. DevOps made some progress as of noon today in applying cicd pipeline fixes. Outstanding items at the time were infrastructure stuff.

Comment by [Brett Roy](#) [13/Oct/25]

[Radhika Kandoori Ana Oliva \(contractor\)](#)

I am walking through these in the QA environment right now ... will contact you later today or early tomorrow to review so you can test

Comment by [Brett Roy](#) [16/Oct/25]

[20251016-qa.7z](#) 

Some QA Stuff

[Ana Oliva \(contractor\)](#) [Radhika Kandoori](#)

Pull this and open before the demo at 2:30 - It will 'hopefully' what I show you!

[Swetha Bayya](#) FYI

Comment by [Brett Roy](#) [22/Oct/25]

[Ana Oliva \(contractor\)](#) [Radhika Kandoori](#)

Fix to Timestamp PR'd - This will go to STAGE and I will need to get it over to QA manually

Will let you know in the AM

Comment by [Brett Roy](#) [23/Oct/25]

[Ana Oliva \(contractor\)](#) [Radhika Kandoori](#)

Fix to timestamp Approved and I see it in QA

It should also be in STAGE

[Jason Thomas](#) looks to have deployed all returns work to STAGE so you should be able to test there when that is ready

In order to test on STAGE, you will need to change the .ps1 script that publishes the text event to point to stage ... the needed change should be obvious

Holler today with any ?s!

cc: [Swetha Bayya](#) [Monika Bhatia](#)

Comment by [Swetha Bayya](#) [23/Oct/25]

[Brett Roy](#) Thank you !

Comment by [Ana Oliva \(contractor\)](#) [24/Oct/25]

<https://wellfit.atlassian.net/servicedesk/customer/portal/5/DIS-984?created=true> waiting for DevOps help with db permissions for Payments db in Stage env

Comment by [Ana Oliva \(contractor\)](#) [28/Oct/25]

Forwarding from slack, waiting for devs help submerchant configs platform/payments db

Hi [@Jason Thomas](#),

I'm having 'submerchant not configured' issue for ACH happy path in stage env, I'm using the one you suggested `'70a4a940-e788-4f1f-b912-fa09e5d7fff4'`

Pointing to that submerchant in:

-> create_test_payment_records.sql

-> create_test_return_configuration.sql

-> publish-stage-ach-returns-from-csv.ps1

However in stage-insights, I get this error message:

Message ReturnNotificationReceivedProcessor - No ReturnConfiguration for SubMerchantId 159f2670-1b71-4ef6-ad30-0ebf0991e2cc. No Redeposit.

Am I missing any other SM config?

Attaching screenshots:

create_test_payment_records.sql - (202) ...it.com

create_test_return_configuration.sql - (100) ...it.com

read_logs.sql - (120) ...it.com

test_returns_2025-09-18.csv

test_returns_2025-09-18.csv

Users > lilu > Downloads > 20251016-stage > create_test_payment_records.sql

Run

Cancel

Disconnect

Change

Database: platform

Estimated Plan

Enable Actual Plan

Parse

Enable SQLCMD

To Notebook

1

-- Path: c:\Source\Wellfit\Database\Platform\Platform\Payments

2

-- Define specific GUIDs for testing

3

DECLARE @PaymentId UNIQUEIDENTIFIER = '12345678-1234-1234-1234-123456789ABC';

4

DECLARE @AchDetailId UNIQUEIDENTIFIER = '98765432-9876-9876-9876-987654321DEF';

5

/* DECLARE @SubMerchantAccountId UNIQUEIDENTIFIER = '12032023-AAAA-BBBB-CCCC-000000000001'; qa */

6

DECLARE @SubMerchantAccountId UNIQUEIDENTIFIER = '70a4a940-e788-4f1f-b912-fa09e5d7ffff'; /*stage*/

7

8

-- SELECTs to View

9

SELECT * FROM [Payments].[Payments] WHERE Id IN (@PaymentId)

10

SELECT * FROM [Payments].[AchPaymentDetails] WHERE PaymentId IN (@PaymentId)

11

SELECT * FROM [Payments].[ReturnedPayments] WHERE OriginalPaymentId IN (@PaymentId)

12

SELECT * FROM [Payments].[SubMerchantAccounts] WHERE Id IN (@SubMerchantAccountId)

13

14

-- Cleanup existing records if they exist

15

DELETE FROM [Payments].[ReturnedPayments] WHERE OriginalPaymentId IN (@PaymentId);

16

DELETE FROM [Payments].[AchPaymentDetails] WHERE PaymentId IN (@PaymentId);

17

DELETE FROM [Payments].[Payments] WHERE Id IN (@PaymentId);

18

19

20

21

IF EXISTS (SELECT 1 FROM [Payments].[AchPaymentDetails] WHERE [PaymentId] = @PaymentId)

22

BEGIN

23

DELETE FROM [Payments].[AchPaymentDetails] WHERE [PaymentId] = @PaymentId;

24

END

Results

Messages

	Id	TimeStamp	Amount	PayFacFee	PaymentType	AccountSuffix	OrderId	TransactionId
1	12345678-1234-1234-1234-123456789abc	2025-10-28 18:01:07.7533108	+00:00	100.00	2.50	ACH	1234	ORDER-TEST-001 8399765698969888

Results grid

	Id	PaymentId	AccountType	SecCode	EntityCreated	EntityUpdated
1	98765432-9876-9876-9876-987654321def	12345678-1234-1234-1234-123456789abc	CHECKING	PPD	2025-10-28 18:01:07.7533108	+00:00

	Id	TimeStamp	DateIssued	Amount	ReasonCode	ReasonDescript...	OriginalPaymen...	SettlementDate	FundingInstruc...	CaseId	FBOFundingIn
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	Id	SubMerchantId	ProcessorId	AccountId	AccountState	EntityUpdated	
1	70a4a940-e788-4f1f-b912-fa09e5d7ffff	159f2670-1b71-4ef6-ad30-0ebf0991e2cc	0	01334267	2	2025-02-13 18:38:51.9700000	+00:00

Ln 1, Col 1 (793 selected) Spaces: 4 UTF-8 CRLF SQL 3 rows MSSQL 00:00:00 stage-platform-wellfit-sqlserver.database.windows.net : platform (202)

Users > lilu > Downloads > 20251016-stage > create_test_return_configuration.sql

Run

Cancel

Disconnect

Change

Database: Payments

Estimated Plan

Enable Actual Plan

Parse

Enable SQLCMD

To Notebook

1

-- Path: c:\Source\Wellfit\Payments\ach-returns-microservice\AchReturns.Database

2

-- Creates a test return configuration record linked to the SubMerchantId from create_test_payment_records.sql

3

4

-- Define specific GUIDs for testing

5

DECLARE @PaymentId UNIQUEIDENTIFIER = '12345678-1234-1234-1234-123456789ABC';

6

DECLARE @ConfigurationId UNIQUEIDENTIFIER = '87654321-4321-4321-4321-987654321ABC';

7

8

/* DECLARE @SubMerchantId UNIQUEIDENTIFIER = '1D550000-3A34-0000-8090-00D43C1753AB'; -- Same as in create_test_payment_records.sql qa */

9

DECLARE @SubMerchantId UNIQUEIDENTIFIER = '70a4a940-e788-4f1f-b912-fa09e5d7ffff'; --stage

10

11

-- SELECTs to View

12

SELECT * FROM [Returns].[RedepositAttempts] WHERE OriginalPaymentId IN (@PaymentId)

13

SELECT * FROM [Returns].[ReturnConfigurations] WHERE Id IN (@ConfigurationId)

14

15

-- Cleanup existing records

16

DELETE FROM [Returns].[RedepositAttempts];

17

DELETE FROM [Returns].[ReturnConfigurations];

18

19

-- Cleanup existing record if it exists

20

IF EXISTS (SELECT 1 FROM [Returns].[ReturnConfigurations]

21

WHERE [SubMerchantId] = @SubMerchantId AND [ProcessorId] = 1)

22

BEGIN

23

DELETE FROM [Returns].[ReturnConfigurations]

24

WHERE [SubMerchantId] = @SubMerchantId AND [ProcessorId] = 1;

25

END

26

27

-- Insert return configuration

28

INSERT INTO [Returns].[ReturnConfigurations]

29

(

30

[Id],

31

[SubMerchantId],

32

[ProcessorId],

33

[MaxRedepositAttempts]

34

)

Results

Messages

	Id	ReturnedPaymen...	OriginalPaymen...	AttemptNumber	RedepositPayme...	RedepositAmount	ScheduledDate	AttemptStatus	ResponseCode	ResponseMessage	ReturnReason
--	----	-------------------	-------------------	---------------	-------------------	-----------------	---------------	---------------	--------------	-----------------	--------------

	Id	SubMerchantId	ProcessorId	MaxRedepositAttempts	EntityCreated	EntityUpdated
1	87654321-4321-4321-4321-987654321abc	70a4a940-e788-4f1f-b912-fa09e5d7ffff	0	2	2025-10-28 18:02:42.8333333	+00:00

Ln 1, Col 1 (609 selected) Spaces: 4 UTF-8 CRLF SQL 1 rows MSSQL 00:00:00 stage-platform-wellfit-sqlserver.database.windows.net : Payments (100)

Users > lilu > Downloads > 20251016-stage > create_test_return_configuration.sql

Run

Cancel

Disconnect

Change

Database: Payments

Estimated Plan

Enable Actual Plan

Parse

Enable SQLCMD

To Notebook

1

-- Path: c:\Source\Wellfit\Payments\ach-returns-microservice\AchReturns.Database

2

-- Creates a test return configuration record linked to the SubMerchantId from create_test_payment_records.sql

3

4

-- Define specific GUIDs for testing

5

DECLARE @PaymentId UNIQUEIDENTIFIER = '12345678-1234-1234-1234-123456789ABC';

6

DECLARE @ConfigurationId UNIQUEIDENTIFIER = '87654321-4321-4321-4321-987654321ABC';

7

8

/* DECLARE @SubMerchantId UNIQUEIDENTIFIER = '1D550000-3A34-0000-8090-00D43C1753AB'; -- Same as in create_test_payment_records.sql qa */

9

DECLARE @SubMerchantId UNIQUEIDENTIFIER = '70a4a940-e788-4f1f-b912-fa09e5d7ffff'; --stage

10

11

-- SELECTs to View

12

SELECT * FROM [Returns].[RedepositAttempts] WHERE OriginalPaymentId IN (@PaymentId)

13

SELECT * FROM [Returns].[ReturnConfigurations] WHERE Id IN (@ConfigurationId)

14

15

-- Cleanup existing records

16

DELETE FROM [Returns].[RedepositAttempts];

17

DELETE FROM [Returns].[ReturnConfigurations];

18

19

-- Cleanup existing record if it exists

20

IF EXISTS (SELECT 1 FROM [Returns].[ReturnConfigurations]

21

WHERE [SubMerchantId] = @SubMerchantId AND [ProcessorId] = 1)

22

BEGIN

23

DELETE FROM [Returns].[ReturnConfigurations]

24

WHERE [SubMerchantId] = @SubMerchantId AND [ProcessorId] = 1;

25

END

26

27

-- Insert return configuration

28

INSERT INTO [Returns].[ReturnConfigurations]

29

(

30

[Id],

31

[SubMerchantId],

32

[ProcessorId],

33

[MaxRedepositAttempts]

34

)

Results

Messages

	Id	ReturnedPaymen...	OriginalPaymen...	AttemptNumber	RedepositPayme...	RedepositAmount	ScheduledDate	AttemptStatus	ResponseCode	ResponseMessage	ReturnReason
--	----	-------------------	-------------------	---------------	-------------------	-----------------	---------------	---------------	--------------	-----------------	--------------

	Id	SubMerchantId	ProcessorId	MaxRedepositAttempts	EntityCreated	EntityUpdated
1	87654321-4321-4321-4321-987654321abc	70a4a940-e788-4f1f-b912-fa09e5d7ffff	0	2	2025-10-28 18:02:42.8333333	+00:00

Ln 1, Col 1 (609 selected) Spaces: 4 UTF-8 CRLF SQL 1 rows MSSQL 00:00:00 stage-platform-wellfit-sqlserver.database.windows.net : Payments (100)

https://wellfit.atlassian.net/si/jira.issueviews:issue-html/PAY-1803/PAY-1803.html

5/6

```
Sent event id=b79bf7f4-5d83-4499-8e28-0c4481493bbd CaseId=TEST-001 (attempt 1)
Done. Sent: 1, Failed: 0
```

Generated at Tue Jul 07 19:31:15 UTC 2026 by Christopher Ariel Valenzuela Medina using Jira 1001.0.0-SNAPSHOT#100292-rev:a7625d93e56786c0b38f2df7db03912499a74edf.